RAMCO VIDYA MANDIR SR. SEC. SCHOOL, THAMARAIKULAM			
CLASS	**:**	**TIME**	**: 3 Hrs**
EXAM NO	**:**	**MARKS**	**: 80**

SECTION A (1 mark each) (16 × 1 = 16)

1) Which of the following is a characteristic feature of ctenophores?

a) Radial symmetry

b) Segmentation

c) Coelom

d) Vertebrae

2) The locomotion of ctenophores is aided by:

a) Siphon

b) Comb plates

c) Fin

d) Spines

3) Question 3

a) The number of veins that carry blood to the heart

b) The number of arteries that carry blood from the heart

c) The presence of hepatic portal system

d) The number of chambers in the heart

4) The eight rows of ciliated comb plates in ctenophores are used primarily for

a) feeding on plankton

b) locomotion

c) protection from predators

d) respiration

5) Ctenophores exhibit both extracellular and intracellular digestion. Which of the following best describes their mode of digestion?

a) Food is ingested and broken down only outside the body cavity

b) Food is ingested and broken down only inside digestive cells

c) Food is first broken down outside the body and then ingested for internal digestion

d) Food is absorbed directly through the skin without ingestion

6) Which of the following is a characteristic feature of Ctenophora?

a) Bilateral symmetry

b) Presence of comb plates

c) Presence of a coelom

d) Having a notochord

7) What type of digestion is performed by Ctenophora?

a) Extracellular only

b) Intracellular only

c) Both extracellular and intracellular

d) Neither

8) Which organelle stores the genetic material of a cell?
a) Ribosome
b) Mitochondrion
c) Chloroplast
d) Nucleus
9) Which of the following structures in ctenophores is responsible for locomotion?
a) Ciliated comb plates
b) Tentacles
c) Bioluminescent cells
d) Choanocytes
10) What unique feature distinguishes ctenophores from other diploblastic organisms?
a) Bioluminescence
b) Radial symmetry
c) Presence of a coelom
d) Intracellular digestion
11) The Golgi apparatus is responsible for
a) DNA replication
b) Protein synthesis
c) Protein modification and sorting
d) Energy production
12) Which organ is not involved in excretion in frogs?
a) Kidney
b) Liver
c) Ureter
d) Cloaca

SECTION B (2 marks each) (5 × 2 = 10)

17) Explain the differences between prokaryotic and eukaryotic cells.
18) Describe the respiratory mechanism of frogs during aquatic and terrestrial phases.
19) Which of the following structures is NOT involved in the nitrogenous waste excretion of a frog? a) cloaca

SECTION C (3 marks each) (7 × 3 = 21)

22) Which of the following structures is NOT a component of a frog's respiratory system? (a) Lungs (b) Skin (c) Liver (d) Buccal cavity
23) Which of the following statements about the structure of chloroplasts is correct? (internal choice: A, B, C, D)
24) Which of the following statements about the stomatogastric nervous system is correct? A) It is present only in insects

25) Which of the following best describes the primary function of mitochondria in eukaryotic cells?

SECTION D (4 marks each) (2 × 4 = 8)

Question 1
Passage
Ctenophores, commonly known as sea walnuts or comb jellies, are exclusively marine, radially symmetrical, diploblastic organisms that exhibit tissue■level organisation. Their bodies are distinguished by eight rows of ciliated comb plates that beat in a coordinated manner to propel the animal through the water. Digestion in ctenophores is both extracellular and intracellular: food is first captured by colloblasts on the comb plates and then partly digested outside the cell before being internalised. Bioluminescence is a prominent feature of many ctenophores, allowing them to produce light by a chemical reaction involving luciferin and luciferase. Reproduction occurs without separate sexes; both eggs and sperm are produced by the same individual and fertilisation is typically external.
a) Describe the structure of the comb plates in ctenophores and explain how they function in locomotion.
b) Discuss the dual mode of digestion in ctenophores and compare it with the digestion of other diploblastic organisms such as sponges and cnidarians.

Question 2
Passage
Bioluminescence is a well■marked property of ctenophores, enabling them to emit light for communication, camouflage, or predator avoidance. The light■producing reaction involves luciferin, luciferase, oxygen, and often ATP, and can be triggered by mechanical stimulation or chemical signalling. Because ctenophores lack separate sexes, reproduction is hermaphroditic: gametes are released into the surrounding water where external

SECTION E (5 marks each) (3 × 5 = 15)

31) Which of the following statements about the reproductive anatomy of the frog is correct?
32) Choose one of the following aspects of ctenophores and describe it in detail: (a) the mechanism of locomotion by comb plates, (b) the dual mode of digestion, (c) the reproduction strategy, (d) the phenomenon of bioluminescence.

ANSWER KEY (SECTION A)

Q1. (a)
Q2. (b)
Q3. (c)
Q4. (b)
Q5. (c)
Q6. (b)
Q7. (c)
Q8. (d)
Q9. (a)
Q10. (a)
Q11. (c)
Q12. (b)